WORKED SOLUTION

STEP I 2006, Question 4

Pure • Elementary geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

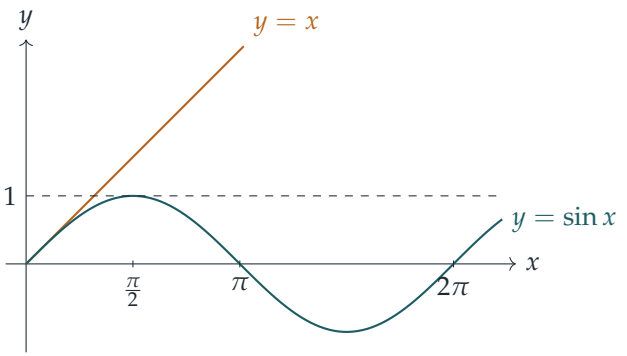
step.maths.org/questions/06-s1-q4

The key idea

The line $y = x$ is the tangent to $y = \sin x$ at the origin, and the sine curve bends away below it. That one picture gives both the inequality and the approximation. For the polygon, split it into n congruent isosceles triangles meeting at the centre; every quantity then becomes trigonometry of the half-angle π/n, and the inequality $\sin x < x$ turns out to be exactly what fixes the sign of the derivative.

Solution

The two graphs



Both curves pass through the origin with gradient 1 there, since $\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x = 1$ at $x = 0$. So $y = x$ is the tangent to the sine curve at O.

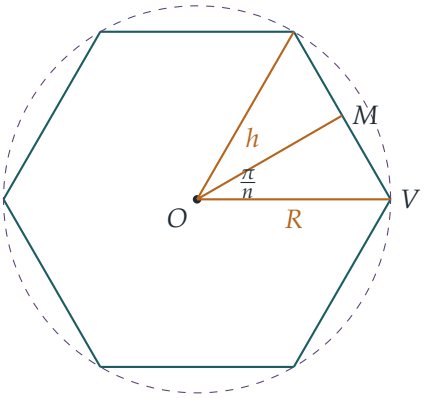
(i) For $0 < x \leq \frac{\pi}{2}$, the sine curve is concave ($\frac{\mathrm{d}^2}{\mathrm{d}x^2}\sin x = -\sin x < 0$), so it lies below its tangent at O, giving $\sin x < x$. (Equivalently: $g(x) = x - \sin x$ has $g(0) = 0$ and $g'(x) = 1 - \cos x > 0$ for $0 < x < 2\pi$, so g is strictly increasing there.) For $x > \frac{\pi}{2}$ we have $\sin x \leq 1 < \frac{\pi}{2} < x$. So $x > \sin x$ for every $x > 0$.

(ii) Because the line is the tangent at O, the two graphs are almost indistinguishable

close to the origin: for small x the height $\sin x$ is very nearly the height x, so $\dfrac{\sin x}{x} \approx 1$. (A sharper statement: since $\sin x \geq x - \frac{x^3}{6}$ for $x > 0$, we have $1 - \frac{x^2}{6} \leq \frac{\sin x}{x} < 1$.)

The area of the polygon

Join the centre O of the polygon to its vertices. This cuts the polygon into n congruent isosceles triangles, each with base equal to a side $s = P/n$ and apex angle $2\pi/n$ at O. Drop the perpendicular from O to the midpoint M of a side; it bisects the apex angle.



In the right-angled triangle OMV (with V a vertex), the angle at O is π/n and $MV = \frac{1}{2}s$. So the distance $h = OM$ from the centre to a side is

$$h = \frac{s/2}{\tan(\pi/n)} = \frac{P}{2n\tan(\pi/n)}.$$

Each of the n triangles has area $\frac{1}{2}sh$, so the polygon has area

$$A = n \cdot \tfrac{1}{2} \cdot \frac{P}{n} \cdot \frac{P}{2n\tan(\pi/n)} = \frac{P^2}{4n\tan(\pi/n)},$$

as required.

The area increases with n

With P fixed, write $A = \dfrac{P^2}{4f(n)}$ where $f(n) = n\tan\dfrac{\pi}{n}$. Since $f(n) > 0$ for $n \geq 3$, A increases exactly when f decreases. By the product and chain rules,

$$f'(n) = \tan\frac{\pi}{n} + n\sec^2\frac{\pi}{n} \cdot \left(-\frac{\pi}{n^2}\right) = \tan\theta - \theta\sec^2\theta, \qquad \text{where } \theta = \frac{\pi}{n}.$$

? **Ask yourself** How do we decide the sign of $\tan\theta - \theta\sec^2\theta$?

Take out the positive factor $\sec^2\theta$ and use $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$:

$$f'(n) = \sec^2\theta\,(\sin\theta\cos\theta - \theta) = \tfrac{1}{2}\sec^2\theta\,(\sin 2\theta - 2\theta).$$

For $n \geq 3$ we have $0 < \theta \leq \frac{\pi}{3}$, so $2\theta > 0$ and part (i) gives $\sin 2\theta < 2\theta$. Hence $f'(n) < 0$, and

$$\frac{\mathrm{d}A}{\mathrm{d}n} = -\frac{P^2 f'(n)}{4f(n)^2} > 0.$$

So, with the perimeter fixed, the area increases as n increases.

Comparison with the circumcircle

The smallest circle containing the polygon is the circumcircle, centred at O through all the vertices. To see that no smaller circle will do, let the vertices be $V_1, \ldots, V_n$ with $\overrightarrow{OV_k} = \mathbf{v}_k$, $|\mathbf{v}_k| = R$. By symmetry $\sum \mathbf{v}_k = \mathbf{0}$, so for any point C with $\overrightarrow{OC} = \mathbf{c}$,

$$\frac{1}{n}\sum_{k=1}^{n} |\mathbf{v}_k - \mathbf{c}|^2 = R^2 - \frac{2}{n}\,\mathbf{c}\cdot\sum_{k=1}^{n}\mathbf{v}_k + |\mathbf{c}|^2 = R^2 + |\mathbf{c}|^2 \geq R^2.$$

So some vertex is at distance at least R from C, and a circle centred at C containing the polygon has radius at least R. The circumradius is $R = OV$, and from the same right-angled triangle,

$$\tfrac{1}{2}s = R\sin\theta \quad\Longrightarrow\quad R = \frac{P}{2n\sin\theta}.$$

So the ratio of the areas is

$$\frac{A}{\pi R^2} = \frac{P^2}{4n\tan\theta}\cdot\frac{4n^2\sin^2\theta}{\pi P^2} = \frac{n}{\pi}\cdot\frac{\sin^2\theta}{\tan\theta} = \frac{\sin\theta\cos\theta}{\theta} = \frac{\sin 2\theta}{2\theta},$$

using $n/\pi = 1/\theta$. For large n, $2\theta = 2\pi/n$ is small, so by part (ii) the ratio is approximately 1.

Checks

For a square ($n = 4$, $\theta = \pi/4$) the formula gives $A = P^2/16$, which is right since each side is $P/4$; the area ratio is $\sin(\pi/2)/(\pi/2) = 2/\pi \approx 0.637$, matching a square of side $\sqrt{2}$ in a unit circle. For a hexagon the ratio is $3\sqrt{3}/(2\pi) \approx 0.827$. We checked the area formula against the shoelace formula for $n = 3, \ldots, 39$, checked $f'(n) < 0$ against a numerical derivative, and found the ratio to be 0.99934 at $n = 100$ and 0.9999934 at $n = 1000$.

Common slips

- Using the full apex angle $2\pi/n$ in the right-angled triangle instead of the half-angle π/n.
- Forgetting the chain-rule factor $-\pi/n^2$ when differentiating $\tan(\pi/n)$.
- Claiming $\tan\theta < \theta\sec^2\theta$ "because $\tan\theta \approx \theta$": an approximation can't settle a sign. The exact inequality $\sin 2\theta < 2\theta$ does.

Beyond STEP

The limit $\sin x / x \to 1$ is the fact that makes the derivative of $\sin x$ equal to $\cos x$, and a first analysis course proves it carefully from a definition of sin rather than from a picture. The monotonicity result is a discrete glimpse of the isoperimetric inequality: among all closed curves of a given length, the circle encloses the most area, and the polygon areas increase towards the circle's value $P^2/(4\pi)$. For a rigorous treatment of limits and derivatives, try Abbott, *Understanding Analysis*.

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